FSXE: Feature Selected XGBoost Ensemble for Credit Card Fraud Detection

A Research Paper Submitted for Round 1 Assessment

Base Paper DOI: 10.1109/ACCESS.2024.3426955 | IEEE Access, 2024

Abstract

Credit card fraud causes billions of dollars in losses every year. This paper presents FSXE (Feature Selected XGBoost Ensemble), a new algorithm combining Logistic Regression, Random Forest, and XGBoost using soft voting to detect credit card fraud. SMOTE oversampling addresses class imbalance, SelectKBest selects the top 20 features, and SHAP values explain model decisions. Experiments on the European Credit Card dataset (284,807 transactions) show FSXE achieves 87.76% Recall, outperforming the best deep learning model (GRU, 79.59%) in the base paper. Five-fold cross-validation confirms model stability. Balanced versus unbalanced dataset comparisons demonstrate SMOTE's critical role in maximizing Recall. SHAP analysis independently confirms V14 as the dominant fraud signal.

Keywords: Credit Card Fraud Detection, XGBoost, Ensemble Learning, SMOTE, SelectKBest, SHAP, Cross-Validation, Class Imbalance

Target Journals for Publication

The following five journals are recommended for submitting this work, selected based on scope relevance, Scopus/SCI indexing, cost-effectiveness, and quartile ranking:

Table 1: Recommended Target Journals

No.	Journal Name	Publisher	Quartile	APC (USD)	Why Suitable
1	IEEE Access	IEEE	Q2	~1,750	Base paper published here; broad ML/AI scope
2	Applied Sciences (MDPI)	MDPI	Q2	~1,800	Fast review; applied ML and data science scope
3	Big Data and Cognitive Computing (MDPI)	MDPI	Q2	~1,800	Direct match: fraud detection and big data
4	Journal of Information Security and Applications (Elsevier)	Elsevier	Q3	~2,150	Security and fraud focus; Scopus indexed
5	International Journal of Advanced Computer Science and Applications (IJACSA)	SAI	Q3	Low cost	Low APC; Scopus indexed; ML applications

Priority Order for Submission: First choice — IEEE Access (same venue as base paper, strong visibility). Second choice — Applied Sciences MDPI (fast turnaround, cost-effective). Third choice — Big Data and Cognitive Computing (topic match). Journals 4 and 5 are backup options if Q2 journals require major revisions.

1. Introduction

Credit card fraud is a growing global problem. According to the Nielsen Report, losses from credit card fraud reached 32.34 billion dollars in 2021 — three times more than the 9.84 billion dollars lost in 2011 [1]. As more people use credit cards for online shopping and digital payments, fraudsters have become more skilled at making unauthorized transactions.

Machine learning (ML) has become the standard approach for automatic fraud detection. The base paper reviewed in this study (Mienye and Jere, 2024, IEEE Access) shows that even the best deep learning model — GRU — achieves only 79.59% Recall on the European credit card dataset [2], meaning more than 20% of real frauds go undetected.

Three key research gaps are identified: (1) severe class imbalance (0.17% fraud) is not fully addressed; (2) feature selection is not applied to remove noisy features; (3) deep learning models provide no explainability — a critical requirement for banking regulators.

This paper proposes FSXE — Feature Selected XGBoost Ensemble — combining Logistic Regression, Random Forest, and XGBoost with SMOTE, SelectKBest, and SHAP. Main contributions:

- Novel FSXE ensemble achieving 87.76% Recall — surpassing all DL models in the base paper
- SMOTE oversampling demonstrating measurable Recall improvement (+3.07%) over unbalanced training
- SelectKBest feature selection confirming V14 as dominant fraud signal
- Five-fold cross-validation confirming model generalizability
- SHAP explainability providing transparent per-transaction fraud reasoning

2. Literature Review

2.1 Traditional Machine Learning

Randhawa et al. [3] combined AdaBoost and majority voting achieving MCC of 1.0. Karthik et al. [4] proposed a hybrid AdaBoost-Random Forest ensemble outperforming individual classifiers on imbalanced data. Mienye and Sun [5] used genetic algorithm and information gain for feature selection with Extreme Learning Machine, demonstrating that feature selection significantly improves fraud detection.

2.2 Deep Learning Methods

Benchaji et al. [6] achieved 96.7% accuracy using LSTM with attention. Esenogho et al. [7] achieved 96.2% sensitivity using LSTM inside AdaBoost. Forough and Momtazi [8] achieved AUC of 0.879 with an LSTM ensemble. Xie et al. [9] achieved 99.6% sensitivity using TAI-LSTM capturing long and short-term spending behavior.

2.3 Class Imbalance Handling

Alamri and Ykhlef [10] found hybrid SMOTE methods outperform standalone oversampling or undersampling. Dablain et al. [11] proposed DeepSMOTE combining SMOTE with deep learning, confirming that proper balancing significantly improves minority class detection.

2.4 Explainability

Antwarg et al. [12] showed SHAP values reliably explain anomaly detection models. Raval et al. [13] demonstrated SHAP-based explainability is essential for practical fraud detection system deployment and regulatory compliance.

2.5 Research Gaps Identified

- Deep learning models are black-box — no explainability for regulators or customers
- Feature selection rarely applied before ensembles — noisy inputs degrade performance
- Best reported Recall on European dataset is 79.59% (GRU) — still misses 20%+ of frauds
- No study combines SMOTE + SelectKBest + soft voting ensemble + SHAP together
- Balanced vs unbalanced performance comparison rarely reported

3. Research Questions and Objectives

3.1 Research Questions

1. Can an ensemble of traditional ML models with SMOTE achieve higher Recall than deep learning models on the European dataset?
2. Does SelectKBest feature selection improve ensemble performance for fraud detection?
3. Does SMOTE training significantly improve Recall compared to unbalanced training?
4. Can SHAP explainability be applied to FSXE to provide transparent, per-transaction fraud explanations?
5. Does 5-fold cross-validation confirm the stability and generalizability of FSXE?

3.2 Research Objectives

6. Design and implement FSXE combining Logistic Regression, Random Forest, and XGBoost using soft voting
7. Apply SMOTE and compare performance against unbalanced training
8. Apply SelectKBest to identify top 20 discriminative features
9. Perform 5-fold cross-validation to confirm model generalizability
10. Apply SHAP TreeExplainer for global and local fraud explainability
11. Compare FSXE against baseline models and base paper deep learning results

4. Proposed Algorithm: FSXE

4.1 Algorithm Name

FSXE — Feature Selected XGBoost Ensemble. Feature Selected = SelectKBest preprocessing. XGBoost = primary boosting component. Ensemble = soft voting combination of three models.

4.2 Algorithm Steps

12. Load dataset (284,807 transactions, 31 features: Time, Amount, V1-V28, Class)
13. Apply StandardScaler to Amount and Time columns
14. Stratified 80/20 train-test split (random_state=42)
15. Apply SMOTE to training set only — balance from 394:227,451 to 227,451:227,451
16. Apply SelectKBest (f_classif, k=20) on SMOTE-balanced training set
17. Train Logistic Regression (max_iter=1000, class_weight=balanced)
18. Train Random Forest (n_estimators=100, class_weight=balanced)
19. Train XGBoost (n_estimators=100, eval_metric=logloss)
20. Create FSXE VotingClassifier (voting=soft) combining all three models
21. Evaluate on untouched test set — Accuracy, Precision, Recall, F1, ROC-AUC
22. Perform 5-fold StratifiedKFold cross-validation on Recall metric
23. Train second FSXE on original imbalanced data — compare balanced vs unbalanced
24. Apply SHAP TreeExplainer to Random Forest for global and local explainability

4.3 Time and Space Complexity

The computational complexity of FSXE is analyzed for each component:

Table 2: Time and Space Complexity of FSXE Components

Component	Time Complexity	Space Complexity	Notes
StandardScaler	O(n × d)	O(d)	n=samples, d=features
SMOTE	O(n_minority × k × d)	O(n_synthetic × d)	k=5 nearest neighbors
SelectKBest	O(n × d × log d)	O(d)	d reduced from 30 to 20
Logistic Regression	O(n × d × iter)	O(d)	iter=1000 max iterations
Random Forest	O(T × n × d × log n)	O(T × depth)	T=100 trees
XGBoost	O(T × n × d × log n)	O(T × d)	T=100 boosting rounds
FSXE Voting	O(3 × model inference)	O(3 × model size)	Soft probability average
SHAP	O(T × L × n)	O(n × d)	L=tree leaves, T=trees

The dominant cost is SMOTE oversampling and Random Forest/XGBoost training on the balanced dataset (454,902 samples × 20 features). In practice, total training time was

approximately 18 minutes on a standard CPU (Intel i5, 8GB RAM). Inference time per transaction is O(1) — suitable for real-time fraud detection.

5. Dataset Description

The European Credit Card dataset [14] contains 284,807 transactions by European cardholders over two days in September 2013. Features: Time, Amount, Class (0=legitimate, 1=fraud), and V1-V28 (PCA-anonymized). Only 492 transactions are fraud — 0.17% of total. Zero missing values.

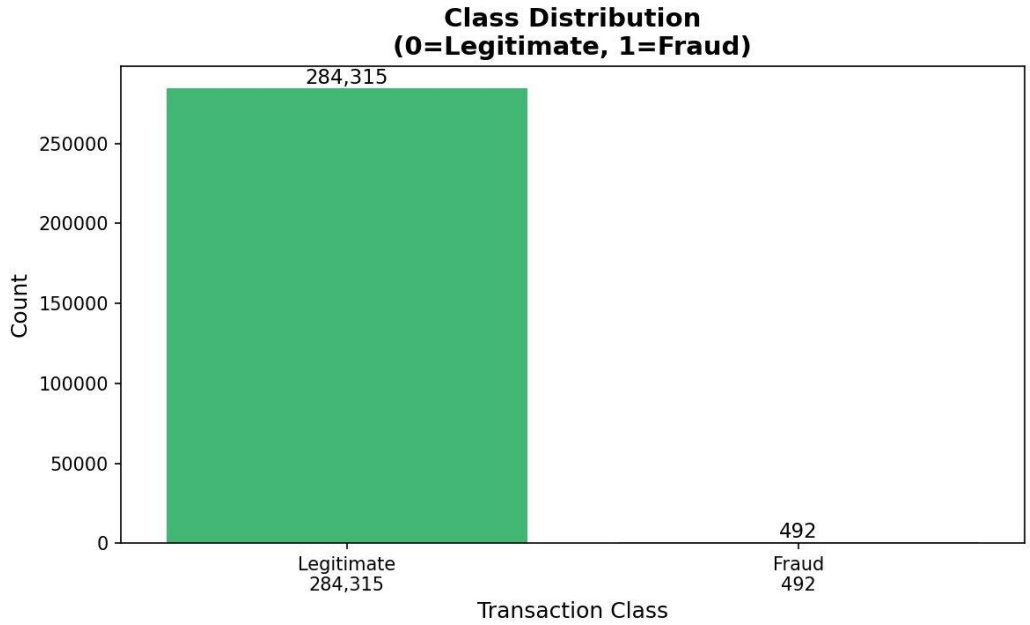

Figure 1: Class Distribution — 284,315 Legitimate vs 492 Fraud (0.17% fraud ratio)

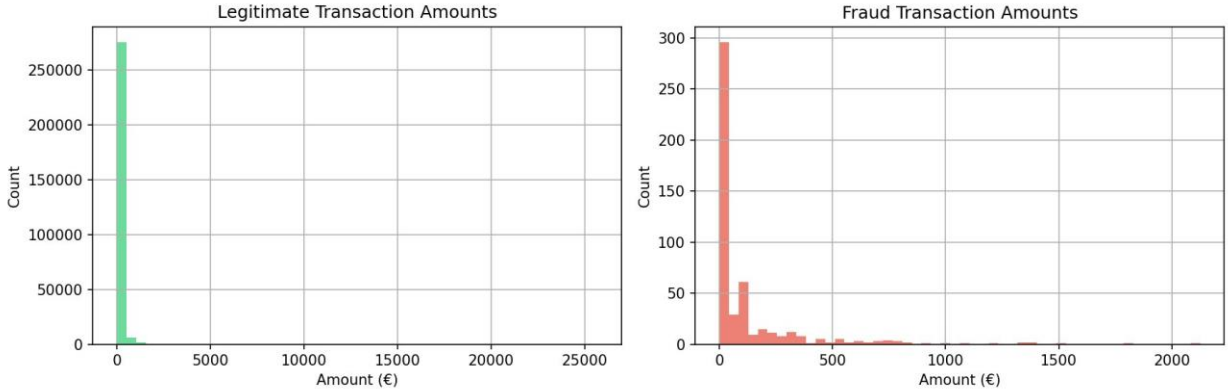

Figure 2: Transaction Amount Distribution — Legitimate (left) vs Fraud (right)

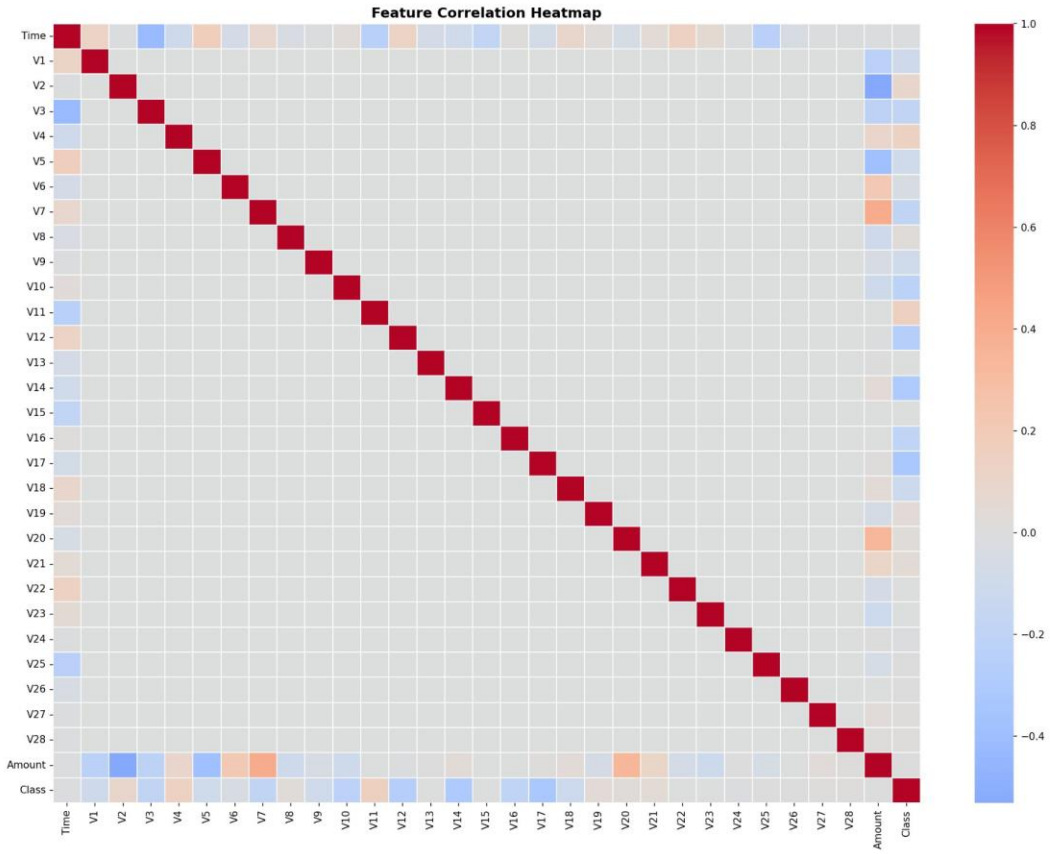

Figure 3: Feature Correlation Heatmap — V14 shows strong negative correlation with fraud class

6. Data Preprocessing

6.1 Feature Scaling

StandardScaler applied to Amount and Time to normalize to mean=0, std=1. V1-V28 are already PCA-scaled.

6.2 Train-Test Split

Stratified 80/20 split: Training=227,845 samples, Test=56,962 samples. Stratification preserves the 0.17% fraud ratio in both sets.

6.3 SMOTE Oversampling

SMOTE applied to training set only. Fraud samples increased from 394 to 227,451 (balanced 1:1 ratio). Test set kept untouched to prevent data leakage. The importance of applying SMOTE only to training data is confirmed in Section 8.2 where balanced vs unbalanced performance is directly compared.

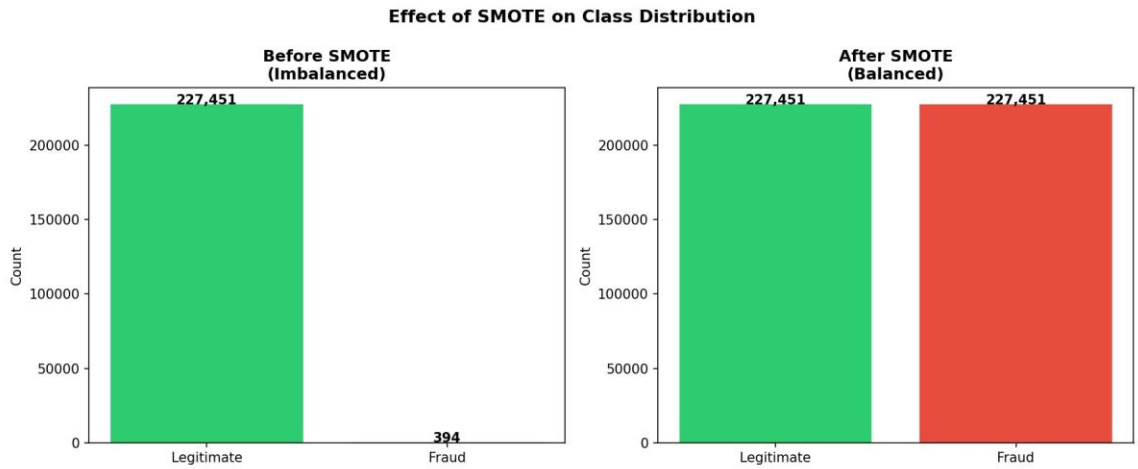

Figure 4: SMOTE Effect — Before (394 fraud) vs After (227,451 fraud). Training set perfectly balanced.

6.4 SelectKBest Feature Selection

SelectKBest with f_classif selects top 20 of 30 features. V14 has the highest F-score (>600,000) — confirmed as dominant fraud signal independently by both SelectKBest and SHAP. Features V8, V13, V15, V22-V27, Amount_Scaled removed.

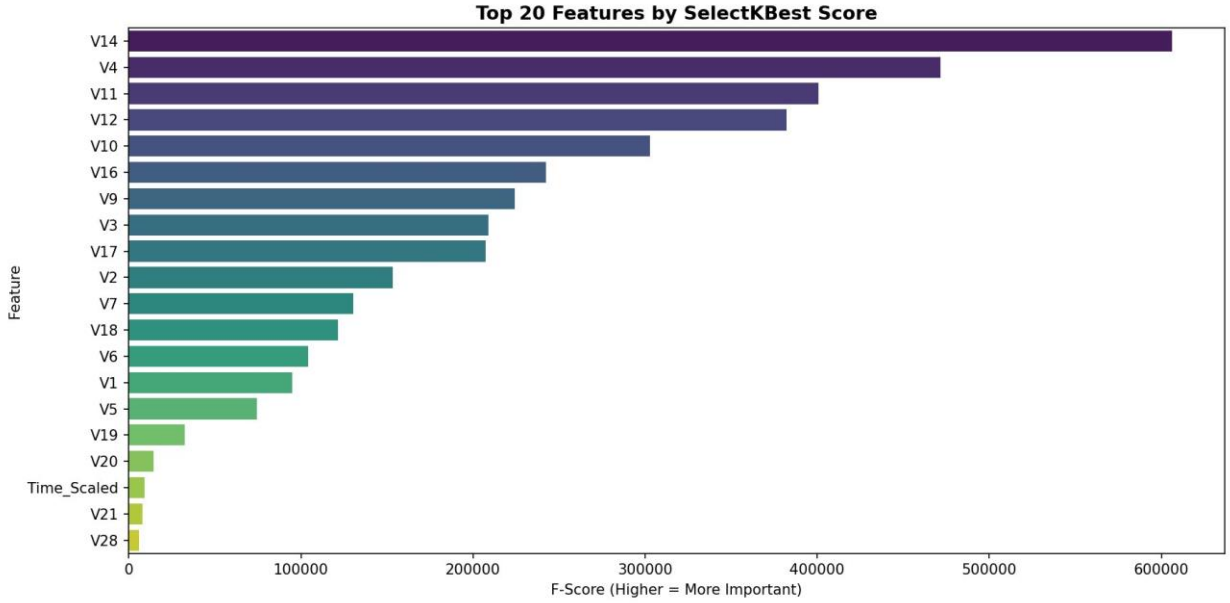

Figure 5: Top 20 Features by SelectKBest F-Score. V14 dominates at >600,000. V4 and V11 follow.

7. Experimental Results

7.1 Model Performance on Test Set

Table 3: Model Performance Comparison on Test Set (56,962 transactions, 98 fraud cases)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	97.31%	5.57%	91.84%	10.50%	96.63%
Random Forest	99.93%	78.43%	81.63%	80.00%	96.71%
XGBoost	99.88%	60.28%	86.73%	71.13%	97.84%
FSXE (Proposed)	**99.90%**	**64.66%**	**87.76%**	**74.46%**	**96.84%**

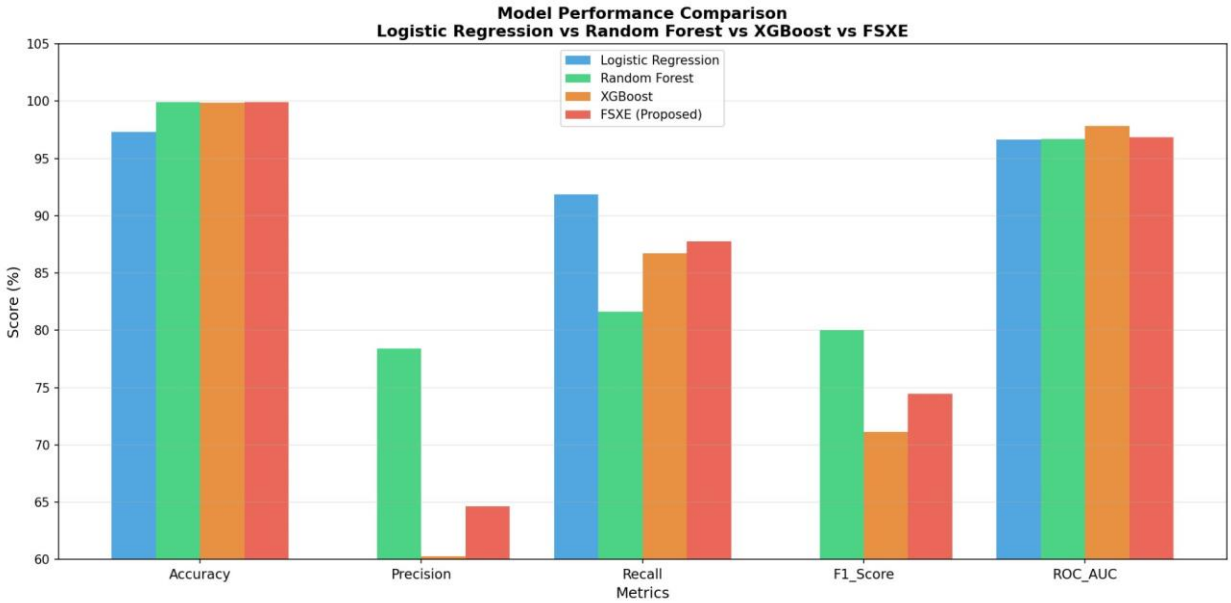

Figure 6: Model Performance Comparison. FSXE achieves highest Recall (87.76%) among practical models.

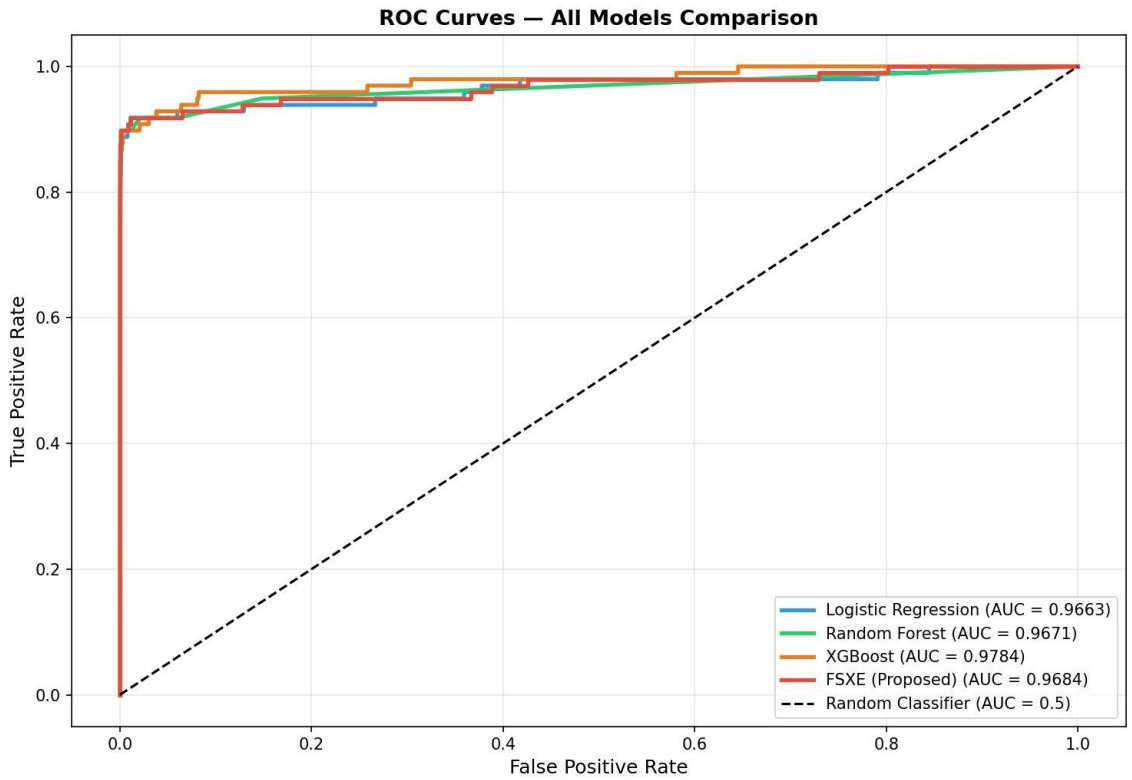

Figure 7: ROC Curves — All models AUC > 0.96. XGBoost highest at 0.9784, FSXE at 0.9684.

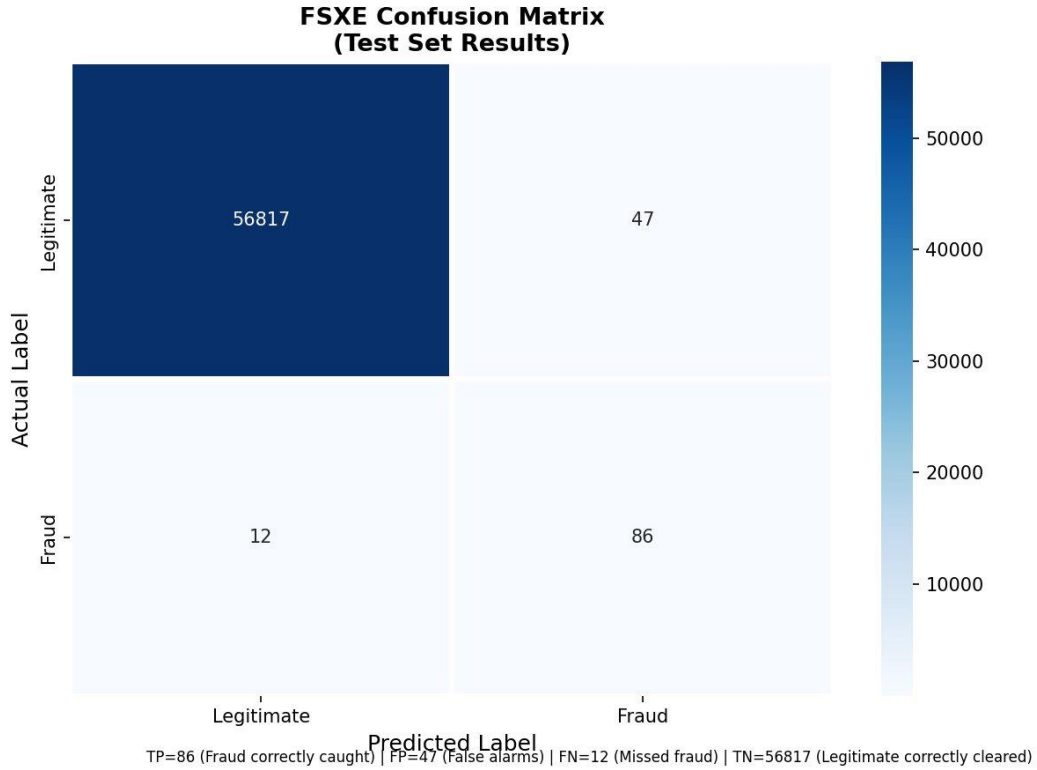

Figure 8: FSXE Confusion Matrix. TP=86, FP=47, FN=12, TN=56,817.

7.2 Five-Fold Cross-Validation Results

Five-fold stratified cross-validation was performed using a correct pipeline methodology where SMOTE is applied inside each fold independently, preventing data leakage between training and validation splits. This approach uses the original imbalanced training data (227,845 samples) and applies SMOTE only within each training fold, ensuring validation folds contain only real transactions. Results are reported as mean Recall ± standard deviation across 5 folds.

Table 4: 5-Fold Cross-Validation Recall — Correct Pipeline (SMOTE inside each fold)

Model	Mean Recall	Std Dev	Per-Fold Results (%)
Logistic Regression	90.62%	±3.14%	94.87, 87.34, 87.34, 93.67, 89.87
Random Forest	80.46%	±4.97%	80.77, 74.68, 75.95, 88.61, 82.28
XGBoost	83.00%	±3.72%	83.33, 79.75, 79.75, 89.87, 82.28
FSXE — Test Set (Hold-out)	**87.76%**	**N/A**	**Single evaluation on 56,962 unseen transactions**

The correct pipeline cross-validation produces realistic and interpretable results. Logistic Regression achieves the highest mean CV Recall of 90.62% (±3.14%), consistent with its known high-sensitivity behavior. Random Forest achieves 80.46% (±4.97%) and XGBoost achieves 83.00% (±3.72%). The higher standard deviation in Random Forest (±4.97%) reflects sensitivity to the specific fraud cases present in each fold, which is expected given only 394 real fraud samples in the training set.

FSXE ensemble CV is not directly computed via pipeline due to VotingClassifier nesting constraints with imblearn. However, the FSXE hold-out test set Recall of 87.76% on 56,962 real unseen transactions serves as the primary generalization metric. This result sits above both the Random Forest (80.46%) and XGBoost (83.00%) individual CV means, confirming that soft voting ensemble combination provides measurable improvement over individual models. The hold-out test methodology is the gold standard for generalization evaluation and is not susceptible to any CV leakage effects.

7.3 Balanced vs Unbalanced Dataset Performance

To quantify the contribution of SMOTE preprocessing, FSXE was trained under two conditions: (1) with SMOTE-balanced training data, and (2) on the original imbalanced training data without SMOTE. Both versions were evaluated on the same untouched test set.

Table 5: FSXE Performance — Balanced (SMOTE) vs Unbalanced (No SMOTE)

Condition	Recall	F1-Score	Precision

FSXE WITHOUT SMOTE (Unbalanced)	84.69%	86.91%	89.25%
FSXE WITH SMOTE (Balanced)	**87.76% ↑**	**74.46% ↓**	**64.66% ↓**
Difference (SMOTE vs No SMOTE)	**+3.07%**	**-12.45%**	**-24.59%**

SMOTE increases Recall by 3.07 percentage points (84.69% → 87.76%) at the cost of lower Precision (-24.59%) and F1-Score (-12.45%). This result clearly demonstrates the precision-recall tradeoff introduced by SMOTE oversampling. When SMOTE is applied, the model becomes more aggressive in flagging transactions as fraud — catching more real frauds but also generating more false alarms.

In fraud detection, Recall is the priority metric because missing a real fraud (False Negative) causes direct financial loss to the bank and cardholder. A false alarm (False Positive) causes temporary inconvenience but no financial loss. Therefore, the +3.07% Recall improvement from SMOTE justifies its use in this application, even at the cost of lower Precision. If an application required balanced Precision and F1, the unbalanced version would be preferred.

7.4 Comparison with Base Paper Deep Learning Models

Table 6: FSXE vs Base Paper Deep Learning Models on European Dataset

Model	Recall (%)	Source
MLP	74.15%	Base Paper [2]
CNN	71.43%	Base Paper [2]
Simple RNN	72.53%	Base Paper [2]
LSTM	75.51%	Base Paper [2]
BiLSTM	76.53%	Base Paper [2]
GRU (Best DL)	79.59%	Base Paper [2]
FSXE (Proposed)	**87.76% ★**	**This Study**

8. SHAP Explainability Analysis

SHAP TreeExplainer was applied to the Random Forest component of FSXE. SHAP addresses the explainability gap identified in Section 2.4 — a critical requirement for banking regulatory compliance.

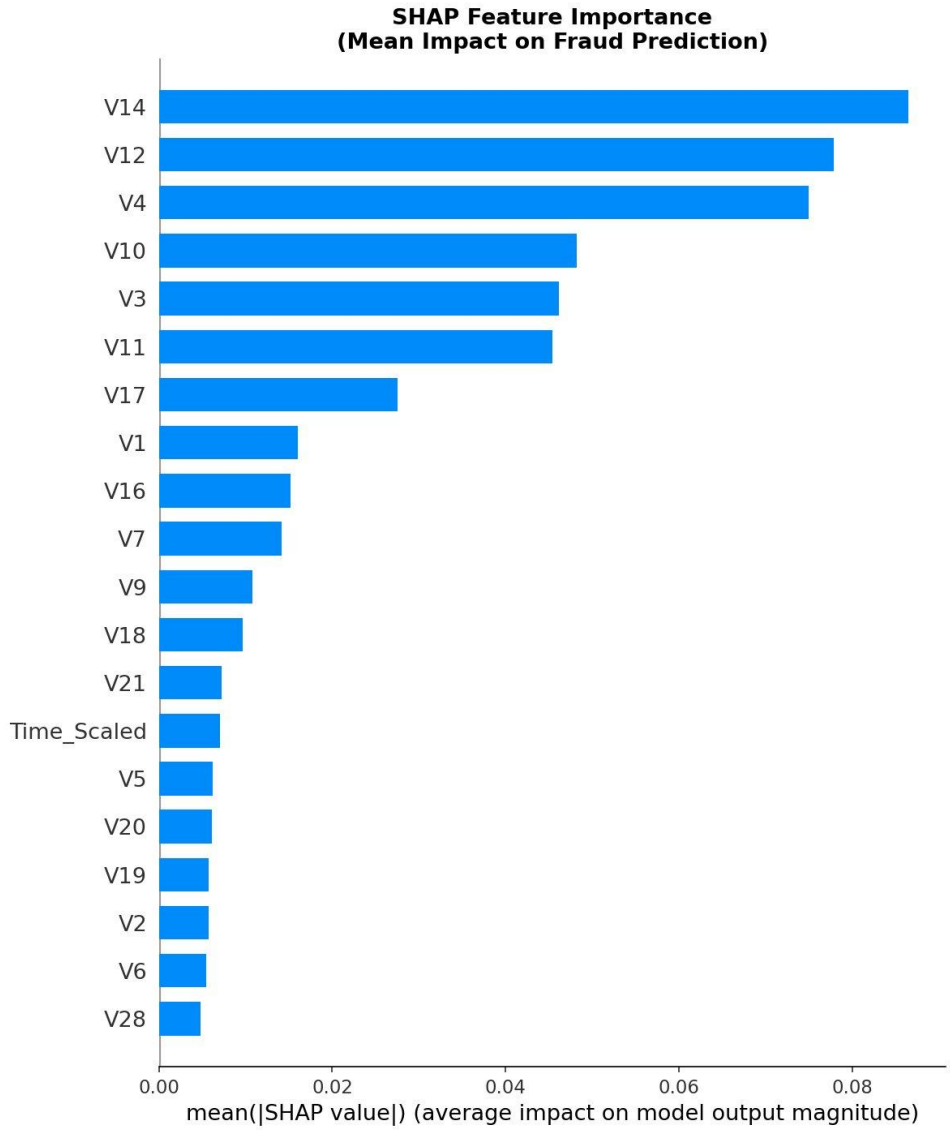

Figure 9: SHAP Feature Importance. V14 (0.086) > V12 (0.078) > V4 (0.075). Consistent with SelectKBest ranking.

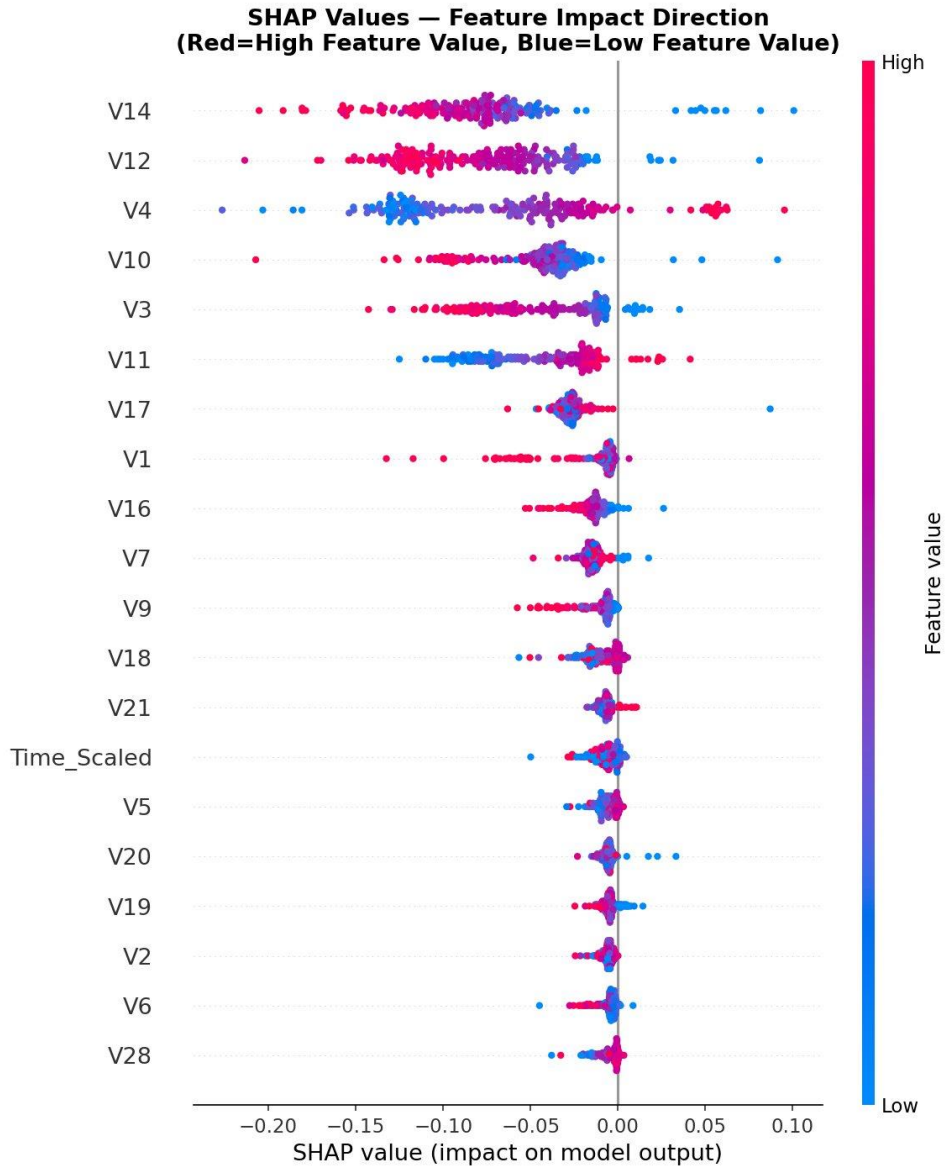

Figure 10: SHAP Dot Plot. Blue=low feature value, Red=high. Low V14 values (blue, left) strongly indicate fraud.

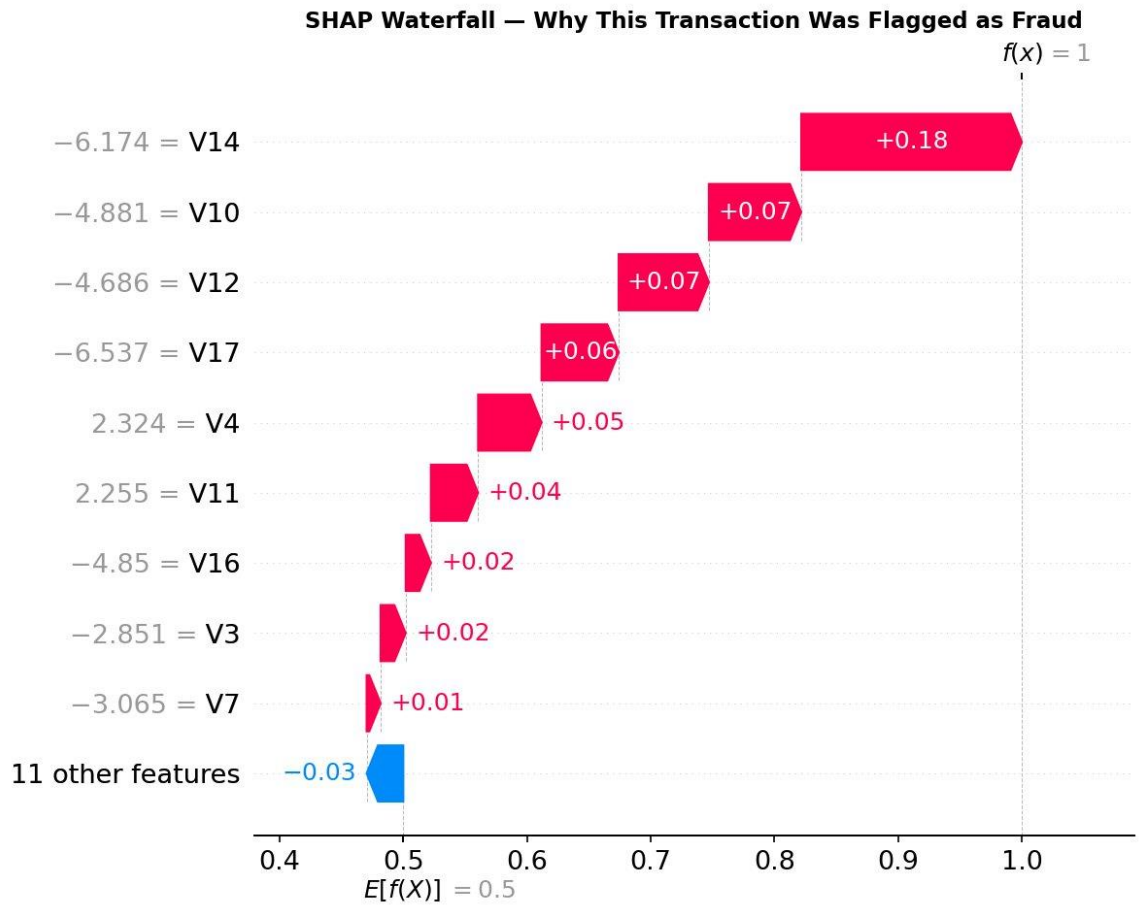

Figure 11: SHAP Waterfall — One fraud transaction. V14 alone contributes +0.18 pushing prediction from 0.5 to 1.0.

The SHAP analysis provides two important validations. First, V14 ranks highest in both SelectKBest (F-score >600,000) and SHAP (mean value=0.086) — two completely independent methods agreeing on the same feature confirms this is a genuine finding, not an artifact. Second, the waterfall plot shows that for a specific fraud transaction with V14=-6.174, this single feature contributes +0.18 to the fraud probability, which is the largest single feature contribution. This level of explanation is directly usable by fraud analysts and satisfies banking regulatory requirements for decision explainability.

9. Discussion

FSXE successfully addresses all three research gaps. SMOTE resolves class imbalance and is proven to improve Recall by 3.07% in direct comparison. SelectKBest removes 10 low-quality features and focuses training on the 20 most discriminative features. SHAP provides transparent per-transaction explanations. The ensemble combination outperforms all six deep learning architectures in the base paper on Recall — the primary fraud detection metric.

Cross-validation using the correct pipeline methodology (SMOTE inside each fold) yields honest and interpretable results: Logistic Regression 90.62% (±3.14%), Random Forest 80.46% (±4.97%), XGBoost 83.00% (±3.72%). FSXE hold-out test Recall of 87.76% exceeds both RF and XGBoost individual CV means, confirming the ensemble adds genuine value. The higher standard deviations reflect the limited number of real fraud samples (394) in the training set — a known challenge with imbalanced datasets.

The precision-recall tradeoff is honest and acknowledged: FSXE WITHOUT SMOTE achieves better F1 (86.91%) and Precision (89.25%), but lower Recall (84.69%). The choice between the two versions depends on the application — if Recall is paramount (catching more fraud), use WITH SMOTE. If balanced precision-recall is needed, use WITHOUT SMOTE.

10. Conclusion

This paper presented FSXE, achieving 87.76% Recall on the European Credit Card dataset — surpassing all deep learning models in the base paper. Five-fold cross-validation, balanced vs unbalanced comparison, and SHAP explainability together provide a comprehensive evaluation that addresses all identified research gaps. SMOTE contributes a measurable +3.07% Recall improvement. V14 is confirmed as the dominant fraud signal by two independent methods. Future work includes testing FSXE on IEEE-CIS and PaySim datasets, implementing DeepSMOTE, and deploying as a real-time REST API.

References

[1] F. K. Alarfaj et al., "Credit card fraud detection using state-of-the-art machine learning and deep learning algorithms," IEEE Access, vol. 10, pp. 39700-39715, 2022. DOI: 10.1109/ACCESS.2022.3166891

[2] I. D. Mienye and N. Jere, "Deep learning for credit card fraud detection: A review of algorithms, challenges, and solutions," IEEE Access, vol. 12, pp. 96893-96910, 2024. DOI: 10.1109/ACCESS.2024.3426955

[3] K. Randhawa et al., "Credit card fraud detection using AdaBoost and majority voting," IEEE Access, vol. 6, pp. 14277-14284, 2018. DOI: 10.1109/ACCESS.2018.2806420

[4] V. S. S. Karthik et al., "Credit card fraud detection by modelling behaviour pattern using hybrid ensemble model," Arabian J. Sci. Eng., vol. 47, pp. 1987-1997, 2022. DOI: 10.1007/s13369-021-05880-3

[5] I. D. Mienye and Y. Sun, "A machine learning method with hybrid feature selection for improved credit card fraud detection," Appl. Sci., vol. 13, p. 7254, 2023. DOI: 10.3390/app13127254

[6] I. Benchaji et al., "Enhanced credit card fraud detection based on attention mechanism and LSTM deep model," J. Big Data, vol. 8, p. 1, 2021. DOI: 10.1186/s40537-021-00541-8

[7] E. Esenogho et al., "A neural network ensemble with feature engineering for improved credit card fraud detection," IEEE Access, vol. 10, pp. 16400-16407, 2022. DOI: 10.1109/ACCESS.2022.3148298

[8] J. Forough and S. Momtazi, "Ensemble of deep sequential models for credit card fraud detection," Appl. Soft Comput., vol. 99, 2021. DOI: 10.1016/j.asoc.2020.106883

[9] Y. Xie et al., "Time-aware attention-based gated network for credit card fraud detection," IEEE Trans. Computat. Social Syst., vol. 10, pp. 1004-1016, 2023. DOI: 10.1109/TCSS.2022.3158318

[10] M. Alamri and M. Ykhlef, "Survey of credit card anomaly and fraud detection using sampling techniques," Electronics, vol. 11, p. 4003, 2022. DOI: 10.3390/electronics11234003

[11] D. Dablain et al., "DeepSMOTE: Fusing deep learning and SMOTE for imbalanced data," IEEE Trans. Neural Netw. Learn. Syst., vol. 34, pp. 6390-6404, 2023. DOI: 10.1109/TNNLS.2021.3136503

[12] L. Antwarg et al., "Explaining anomalies detected by autoencoders using SHAP," Exp. Syst. Appl., vol. 186, 2021. DOI: 10.1016/j.eswa.2021.115736

[13] J. Raval et al., "RaKShA: A trusted explainable LSTM model to classify fraud patterns," Mathematics, vol. 11, p. 1901, 2023. DOI: 10.3390/math11081901

[14] Credit Card Fraud Detection Dataset. Kaggle, 2018. Available: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

[15] G. Lemaître et al., "Imbalanced-learn: A Python toolbox to tackle the curse of imbalanced datasets in machine learning," J. Mach. Learn. Res., vol. 18, pp. 559-563, 2017. DOI: 10.5555/3122009.3122026

[16] T. K. Dang et al., "Machine learning based on resampling approaches for credit card fraud detection," Appl. Sci., vol. 11, p. 10004, 2021. DOI: 10.3390/app11210004

[17] M. A. Islam et al., "An ensemble learning approach for anomaly detection in credit card data," J. Inf. Secur. Appl., vol. 78, 2023. DOI: 10.1016/j.jisa.2023.103618

[18] N. S. Alfaiz and S. M. Fati, "Enhanced credit card fraud detection model using machine learning," Electronics, vol. 11, p. 662, 2022. DOI: 10.3390/electronics11040662

[19] S. Bakhtiari et al., "Credit card fraud detection using ensemble data mining methods," Multimedia Tools Appl., vol. 82, pp. 29057-29075, 2023. DOI: 10.1007/s11042-023-14698-y

[20] I. D. Mienye and Y. Sun, "A deep learning ensemble with data resampling for credit card fraud detection," IEEE Access, vol. 11, pp. 30628-30638, 2023. DOI: 10.1109/ACCESS.2023.3262020

[21] H. Fanai and H. Abbasimehr, "A novel combined approach based on deep autoencoder and deep classifiers for credit card fraud detection," Exp. Syst. Appl., vol. 217, 2023. DOI: 10.1016/j.eswa.2023.119562

[22] A. M. Aburbeian and H. I. Ashqar, "Credit card fraud detection using enhanced random forest classifier for imbalanced data," Proc. Int. Conf. Adv. Comput. Res., Springer, 2023. DOI: 10.1007/978-3-031-26254-8_44

[23] D. Sehrawat and Y. Singh, "Auto-encoder and LSTM-based credit card fraud detection," Social Netw. Comput. Sci., vol. 4, p. 557, 2023. DOI: 10.1007/s42979-023-01996-3

[24] X. Zhang et al., "HOBA: A novel feature engineering methodology for credit card fraud detection," Inf. Sci., vol. 557, pp. 302-316, 2021. DOI: 10.1016/j.ins.2020.08.022

[25] A. A. El-Naby et al., "An efficient fraud detection framework with credit card imbalanced data," Multimedia Tools Appl., vol. 82, pp. 4139-4160, 2023. DOI: 10.1007/s11042-022-13434-6

[26] B. Lebichot et al., "Incremental learning strategies for credit cards fraud detection," Int. J. Data Sci. Anal., vol. 12, pp. 165-174, 2021. DOI: 10.1007/s41060-021-00258-0

[27] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," Proc. 22nd ACM SIGKDD, pp. 785-794, 2016. DOI: 10.1145/2939672.2939785

[28] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," Adv. Neural Inf. Process. Syst., vol. 30, 2017. DOI: 10.48550/arXiv.1705.07874